Question 4: Implement centralized logging using the ELK stack and demonstrate log collection, parsing, and visualization in Kibana.

The ELK stack (Elasticsearch 8.14, Logstash, Kibana) was deployed with Docker Compose. Logstash collected the application log file, grok-parsed it into structured fields and indexed 765+ documents, visualized in Kibana Discover.

Logstash pipeline (logging/logstash/pipeline/app.conf)

```
input {
  file {
    path => "/var/log/lab3/app.log"
    start_position => "beginning"
    sincedb_path => "/dev/null"
  }
}

filter {
  grok {
    match => { "message" => "%{TIMESTAMP_ISO8601:log_time} %{LOGLEVEL:level} %{GREEDYDATA:log_message}" }
  }
  date {
    match => ["log_time", "ISO8601"]
    target => "@timestamp"
  }
  mutate { remove_field => ["message", "log_time", "host", "path", "@version"] }
}

output {
  elasticsearch { hosts => ["http://elasticsearch:9200"] index => "lab3-app-logs" }
}
```

Deployment and verification

```
$ sudo sysctl -w vm.max_map_count=262144   # elasticsearch requirement
$ docker compose -f logging/docker-compose.yml up -d
lab3-elasticsearch Started   lab3-logstash Started   lab3-kibana Started

$ curl http://localhost:9200/_cluster/health
{"status":"yellow", ...}   # single node: replicas unassigned

$ curl -s "http://localhost:9200/_cat/indices/lab3-app-logs?v"
yellow open lab3-app-logs  1 1  765  0  225.9kb

$ sample parsed document:
"level": "INFO",
"log_message": "GET /metrics served",
"@timestamp": "2026-11-04T20:58:..."
```

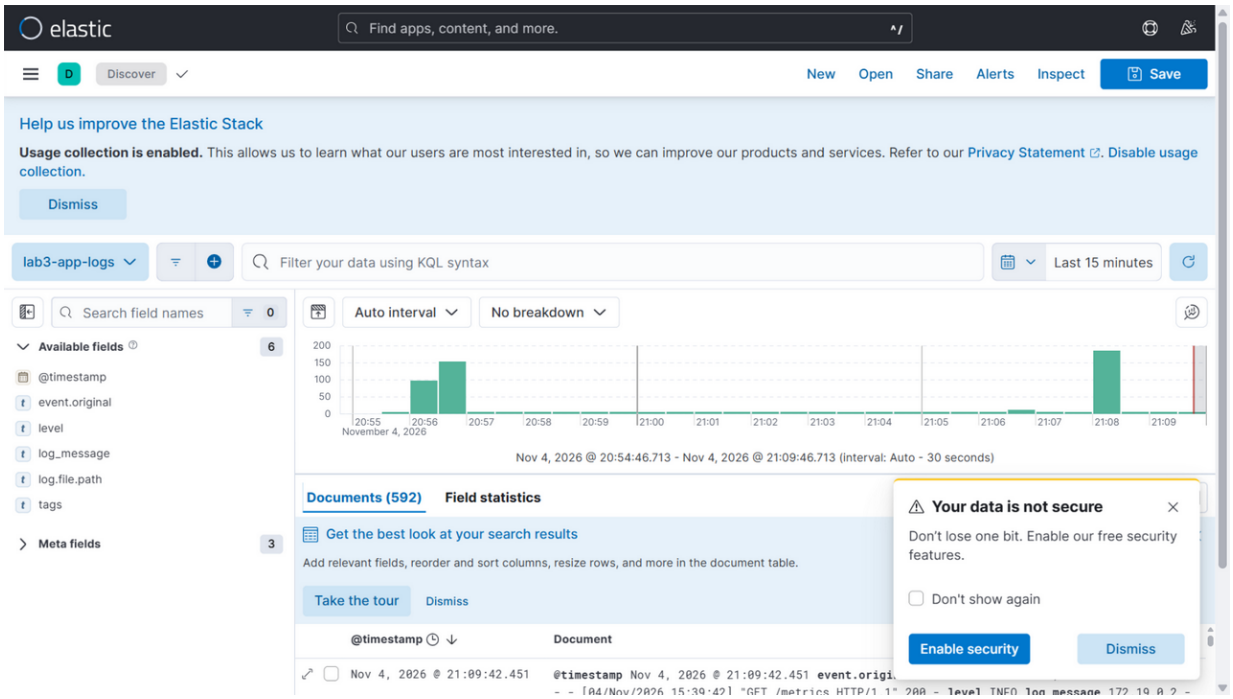


Fig: Kibana Discover: 592+ documents of parsed application logs

Observations

- Log flow: app writes a log file on a shared volume -> Logstash file input -> grok parse -> Elasticsearch index -> Kibana visualizes.
- Grok turned raw lines into structured fields (level, log_message, @timestamp) enabling filters.
- Elasticsearch in Docker requires vm.max_map_count=262144 on the host (WSL2 default is 65530).
- Cluster health yellow is expected on a single node: the replica shard cannot be placed on the same node. xpack.security was intentionally disabled for the lab (Kibana banner says so).

Conclusion

All experiments were performed on Ubuntu 24.04 LTS with Ansible, Terraform, Prometheus, Grafana, the ELK stack and Selenium. The evidence (command logs, configuration files, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-3 under the docs/ directory.